

SYDNEY GRAMMAR SCHOOL



2014 FORM V ANNUAL EXAMINATION

Monday 1st September, 12:55 pm

Chemistry

General Instructions

- Working time – 2 hours
- Board-approved calculators may be used
- Write using blue or black pen
- Draw diagrams using pencil
- A Data Sheet and Periodic Table are provided at the back of this paper
- **Write your name and Master's initials at the top of each page in Part B**
- **Remove the central staple before handing in paper**

Total marks (87)

This paper consists of two parts, **Part A** and **Part B**.

Part A

Total marks (13)

- Attempt ALL Questions
- Allow about 15 minutes for this Part.

Part B

Total marks (74)

- Attempt ALL questions
- Allow about 1 hour and 45 minutes for this Part.

CHECKLIST

Each boy should have the following:

1 Question Paper

1 Multiple Choice Answer Sheet

Chemistry Classes

5CY201 - AKBB	5CY202 - CF	5CY203 - ASG	
5CY204 - EJS	5CY205 - MRB	5CY206 - TW	5CY207 - ZI

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Part A

Total marks (13)

Attempt ALL Questions

Allow about 20 minutes for this Part

Use the multiple-choice Answer Sheet.

Select the alternative A, B, C or D that best answers the question. Fill the response circle completely.

Sample $2 + 4 =$

(A) 2

(B) 6

(C) 8

(D) 9

A

B

©

Ⓓ

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A



©

Ⓓ

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.



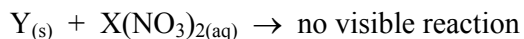
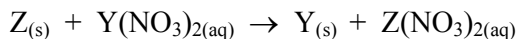
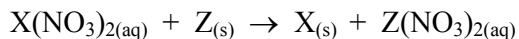
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D

correct

- 1 Which of the following methods would best enable you to separate two immiscible liquids?
- (A) using a separating funnel
 - (B) using a filter funnel
 - (C) using evaporation
 - (D) using distillation
- 2 Which compound contains both covalent and ionic bonds?
- (A) MgBr_2
 - (B) CHCl_3
 - (C) BCl_3
 - (D) NH_4NO_3
- 3 An atom of element Z (not correct symbol) has the electronic configuration 2, 8, 6. Which of the following species is this element most likely to form?
- (A) the ion Z^{2+}
 - (B) the ion Z^{6+}
 - (C) the compound H_2Z
 - (D) the compound Z_6F
- 4 The first four successive ionisation energies (measured in kJ mol^{-1}) for an element are 550, 1064, 4210 and 5500. This element should be placed in the same group as:
- (A) lithium
 - (B) beryllium
 - (C) boron
 - (D) carbon
- 5 Topaz is a mineral with the formula $\text{Al}_2\text{SiO}_4(\text{OH})_2$. What is the percentage by mass of aluminium in topaz?
- (A) 15%
 - (B) 16%
 - (C) 30%
 - (D) 45%

- 6 A boy was investigating the reactivity of three unknown metals (X, Y and Z) using displacement reactions. He recorded the following balanced chemical equations:



Which of the following correctly places the three metals in order of **increasing** reactivity?

- (A) Y, X, Z
 - (B) Z, Y, X
 - (C) X, Z, Y
 - (D) Z, X, Y
- 7 Which of the following best describes the molecular shape of methane?
- (A) trigonal planar
 - (B) bent
 - (C) trigonal pyramidal
 - (D) tetrahedral
- 8 In which of the following substances would you expect hydrogen bonding to occur?

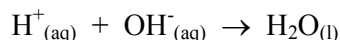
- I. CH_2F_2
- II. CH_3COOH
- III. $\text{C}_2\text{H}_5\text{OH}$

- (A) II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

- 9 The specific heat capacity of gold is $0.128 \text{ J g}^{-1} \text{ K}^{-1}$, and the specific heat capacity of copper is $0.385 \text{ J g}^{-1} \text{ K}^{-1}$. Ten-gram samples of each metal were heated to 60°C , and then dropped into separate beakers containing 20 mL of water.

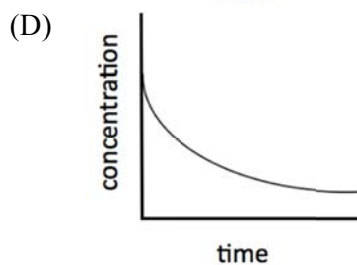
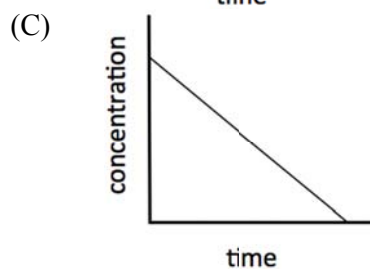
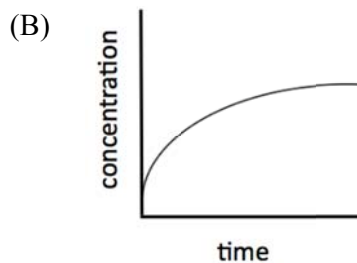
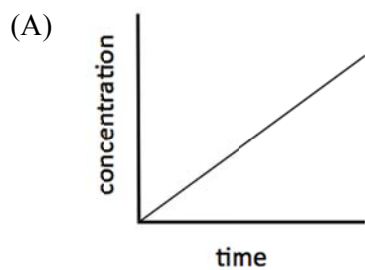
Which of the following statements are correct?

- (A) The copper will react with the water, but the gold will not.
 - (B) The copper sample will transfer more heat energy to the water than the gold sample.
 - (C) The water with the gold sample will have has a final temperature three times higher than the water with the copper sample.
 - (D) The water in the two beakers will remain at the same temperature at all times.
- 10 Wicks are used in candles to:
- (A) Decrease the ignition temperature of the wax vapour.
 - (B) Facilitate the evaporation of the liquid wax.
 - (C) Prevent the whole candle surface from combusting.
 - (D) To attract enough heat from the rest of the candle to liquefy the wax.
- 11 When aqueous solutions of hydrochloric acid and sodium hydroxide are mixed, the temperature of the resulting solution increases. Therefore, the reaction



- (A) is endothermic with $\Delta H^\circ > 0$
 - (B) is endothermic with $\Delta H^\circ < 0$
 - (C) is exothermic with $\Delta H^\circ < 0$
 - (D) is exothermic with $\Delta H^\circ > 0$
- 12 The energy obtained from coal is derived from:
- (A) The compressional energy absorbed by the coal from overlying rocks when buried deep underground.
 - (B) Heat energy absorbed from nearby geothermal sites.
 - (C) Nuclear energy absorbed by the coal from transuranic elements in the Earth's core.
 - (D) Light energy from the Sun absorbed by plants before they were buried and became coal.

- 13 Which of the following graphs best represent the change in concentration of products over time for a reaction as it goes towards completion?



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Part B**Total marks (74)****Attempt ALL Questions**

Master's initials

Name

Answer the questions in the spaces provided.

Show **all** relevant working in questions involving calculations.**Question 14** (2 marks)**Marks**

Write word equations for;

- (a) the electrolysis of water.

1

- (b) photosynthesis.

1**Question 15** (3 marks)

Write balanced chemical equations for;

- (a) the decomposition of silver chloride.

1

- (b) the reaction between solutions of barium hydroxide and copper(II) sulfate.

2

Question 16 (3 marks)**Marks**

Draw Lewis electron dot diagrams for each of the following substances.

(a) hydrogen sulfide

1

(b) nitrogen trifluoride

1

(c) ethene

1

Question 17 (4 marks)**Marks**

The table below summarises some properties of Substance **Y** and Substance **Z**. Use the information to answer the following questions.

Substance Y	Substance Z
white solid that melts at 801 °C to form colourless liquid	grey solid that melts at 419 °C and boils at 906 °C
does not conduct electricity in solid state, but is a good conductor when liquid	conducts electricity in both solid and liquid states

- (a) Describe the structure and bonding of Substance **Z** at 25 °C.

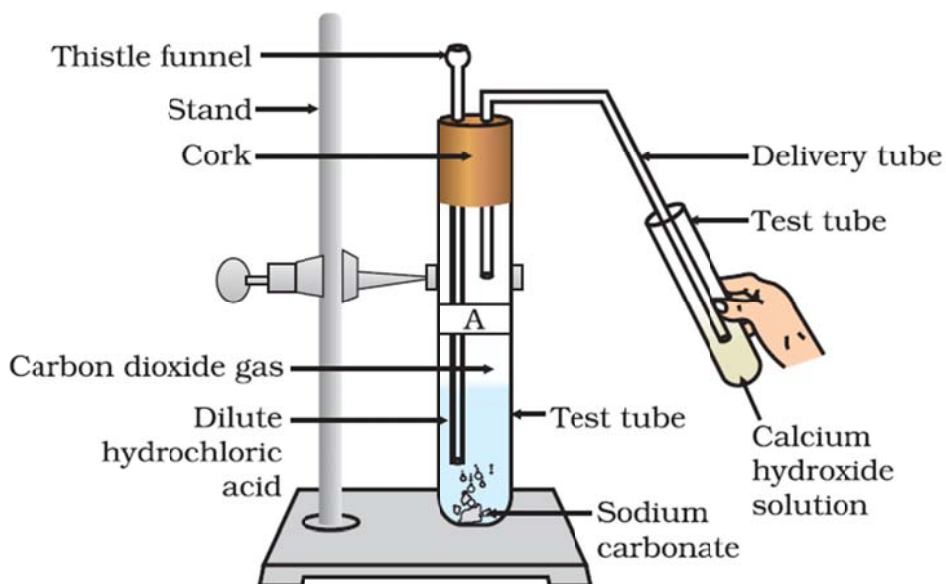
2

- (b) Predict whether an aqueous solution of Substance **Y** would conduct electricity, and justify your answer in terms of the properties of Substance **Y**.

2

Question 18 (4 marks)**Marks**

The diagram below shows an experimental set up for the production of carbon dioxide.



- (a) Write a balanced chemical equation for the reaction producing CO_2 .

1

- (b) If 2.50 g of solid sodium carbonate was reacted with excess hydrochloric acid, calculate the **mass and volume** of carbon dioxide evolved, measured at 100 kPa and 25 °C.

3

Question 19 (3 marks)

Explain why different isotopes of an element have the same chemical properties, but different relative atomic masses.

3

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Master's initials

Name

Question 20 (3 marks)**Marks**

The history of metal use has been intricately interwoven with the evolution of society and civilisations throughout our history. Explain, citing examples, the general trend in technological developments and access to energy with regards to mankind's ability to extract metals from their ore.

3

Question 21 (5 marks)**Marks**

The compound uracil is one of the bases found in RNA (ribonucleic acid). A 336.3 g sample of uracil was broken down into its constituent elements and found to contain 144.0 g of carbon, 12.1 g of hydrogen, 85.1 g of nitrogen, with the remaining mass being oxygen.

- (a) Calculate the molar ratio of the elements in this compound and thus state its empirical formula.

3

- (b) If one mole of uracil contains 28.02 g of nitrogen, calculate the molecular formula of this compound.

2

Question 22 (6 marks)**Marks**

The metal activity series can be experimentally derived using the reactions of metals with acids.

- (a) Redox reactions involve the transfer of electrons. Use a balanced chemical equation and the relevant half equations to explain why the reaction of zinc with aqueous sulfuric acid can be considered to be a redox reaction.

3

- (b) Not all metals react with acids. Explain why some metals will react with an acid (such as aqueous sulfuric acid), whilst others will not.

3

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Master's initials

Name

Question 23 (2 marks)**Marks**

Mendeleev is universally acknowledged as having put together the first functioning Periodic Table. Outline two decisions he made which allowed him to succeed in putting together a functioning Periodic Table when others had failed.

2

Marks**Question 24** (5 marks)

The boiling points for Group V hydrides are shown in the table below.

Hydride	Boiling point (°C)
ammonia (NH ₃)	-33
phosphine (PH ₃)	-88
arsine (AsH ₃)	-55
stibine (SbH ₃)	-17
bismuthine (BiH ₃)	22

- (a) Explain the general trend in boiling points observed for phosphine to bismuthine.

2

- (b) Using a diagram, explain why the boiling point of ammonia does not follow the same general trend as phosphine to bismuthine.

3

Question 25 (5 marks)**Marks**

At 25 °C, the maximum amount of sodium chloride that will dissolve in 100 mL of water is 35.9 g.

- (a) What is the concentration (in mol L⁻¹) of a saturated solution of sodium chloride at 25 °C?

2

- (b) 25.0 mL of a saturated solution of sodium chloride is diluted to 300 mL. Calculate the concentration (in mol L⁻¹) of the final solution.

1

- (c) 40.0 g of sodium chloride is added to 100 mL of deionised water at 25 °C. Using this solution as an example, describe the movement of solute particles in a saturated solution.

2

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Master's initials

Name _____

Marks

Question 27 (3 marks)**Marks**

The image below shows a short “thread” of water suspended between two beakers.



Explain why it is possible to produce a stable “thread” of water as shown above.

3

Question 28 (2 marks)

A hydrocarbon will burn completely once its ignition temperature is reached, even though only a fraction of its molecules will have enough energy to react. Explain this observation.

2**Question 29** (3 marks)

Analyse the impact of increasing the temperature on reaction rate.

3

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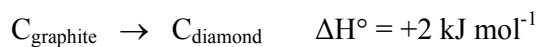
Master's initialsName**Question 30** (6 marks)**Marks**

Carbon has a number of different allotropic forms, including diamond and graphite.

- (a) Compare the nature and geometry of the bonding in diamond and graphite.

2

- (b) Under the right conditions, graphite can be converted into diamond.



The activation energy for this process is 120 kJ mol^{-1} .

- (i) Outline why the activation energy for this process is so high.

1

- (ii) Given the above information, suggest what reaction conditions would be required to convert graphite into diamond.

1

Question 30 continued on next page.

Question 30 continued.**Marks**

- (ii) Draw a fully labelled energy profile diagram for the reaction transforming graphite to diamond.

2**Question 31 (4 marks)**

Describe a reliable experiment that you could perform in the laboratory, using magnesium and hydrochloric acid, to investigate the effect of changing concentration on reaction rate. Include the expected result of this experiment.

4

Question 32 (3 marks)**Marks**

A BBQ heat bead, comprised of a mixture of graphite and clay, was placed in a container. An excess of oxygen gas was passed over the heat bead as it was being combusted. The graphite underwent complete combustion, but the clay remained unchanged. Masses of the container and its contents were documented during the experiment along with the mass of CO₂ produced.

Determine the percentage by mass of graphite in the BBQ heat bead using the information in the table below.

Item	Mass (kg)
reaction container	2.45
reaction container with heat beads before combustion	3.75
evolved CO ₂	2.89

3

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Master's initialsName**Question 33** (4 marks)**Marks**

The following information refers to the reactions of an unknown compound, Compound **Y**. Compound **Y** is a molecular gas composed of nitrogen and oxygen. All gas volumes are measured at 25 °C and 100 kPa.

4

- Compound **Y** can be prepared by the reaction of another gas, Compound **X**, with oxygen. In this reaction, 1.00 L of Compound **X** reacts with 0.500 L of oxygen to produce 1.00 L of Compound **Y**.
- Compound **Y** will react with lead(II) oxide to produce Compound **X** and a white solid, Compound **Z**. In this reaction, 3.00 g of lead(II) oxide will react with 1.00 L of Compound **Y**, to produce 4.45 g of Compound **Z** and 0.333 L of Compound **X**. Compound **Z** is soluble in water.
- Compound **Y** has the interesting property that it can be used as a source of oxygen to promote combustion reactions. For example, it can be used as the oxygen source for the complete combustion of hydrocarbons, so that oxygen gas does not need to be used.

Showing all necessary logic, write balanced chemical equations for the three chemical reactions below. Ensure you clearly identify the formulae of Compounds **X**, **Y** and **Z**;

- a) The reaction of Compound **X** with oxygen to produce Compound **Y**.
- b) The reaction of Compound **Y** with lead(II) oxide to produce Compound **Z** and Compound **X**.
- c) The reaction of methane and Compound **Y**.

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Chemistry**Data Sheet**

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0 °C (273 K)	22.71 L
at 25 °C (298 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \quad \Delta H = -mC\Delta T$$

Standard Potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}_{(\text{s})}$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}_{(\text{s})}$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}_{(\text{s})}$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}_{(\text{s})}$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}_{(\text{s})}$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}_{(\text{s})}$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}_{(\text{s})}$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2} \text{H}_{2(\text{g})} + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}_{(\text{s})}$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}_{(\text{s})}$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}_{(\text{s})}$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}_{(\text{s})}$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}_{(\text{s})}$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2} \text{H}_{2(\text{g})}$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_{2(\text{g})} + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}_{(\text{s})}$	0.34 V
$\frac{1}{2} \text{O}_{2(\text{g})} + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}_{(\text{s})}$	0.52 V
$\frac{1}{2} \text{I}_{2(\text{s})} + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2} \text{I}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}_{(\text{s})}$	0.80 V
$\frac{1}{2} \text{Br}_{2(\text{l})} + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2} \text{Br}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2} \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2} \text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2} \text{H}_2\text{O}$	1.36 V
$\frac{1}{2} \text{Cl}_{2(\text{g})} + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2} \text{Cl}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2} \text{F}_{2(\text{g})} + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

KEY

PERIODIC TABLE OF THE ELEMENTS																	
1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium					
		Atomic Number Symbol Standard Atomic Weight Name															
3 Li 6.941 Lithium	4 Be 9.012 Beryllium											5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium											13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	35 Br 79.90 Bromine	36 Kr 83.80 Krypton	53 I 126.9 Iodine	54 Xe 131.3 Xenon		
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc 98.00 Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	52 Te 127.6 Tellurium	54 Xe 131.3 Xenon	85 At 210.0 Astatine	86 Rn 222.0 Radon		
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57-71 Lanthanoids		72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	82 Pb 207.2 Lead	84 Po 209.0 Polonium	85 At 210.0 Astatine	86 Rn 222.0 Radon		
87 Fr 223.0 Francium	88 Ra 226.0 Radium	89-103 Actinoids		104 Rf 261.0 Rutherfordium	105 Db 262.0 Dubnium	106 Sg 266.0 Seaborgium	107 Bh 264.0 Bohrium	109 Mt 268.0 Meitnerium	110 Ds 271.0 Darmstadtium	111 Rg 272.0 Roentgenium	112 Cn 285.0 Copernicium						

Lanthanoids

57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
138.9	140.1	140.9	144.2		150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.1	175.0

Actinoids

89	Th Ac
90	Th Thorium
91	Pa Protactinium
92	U Uranium
93	Np Neptunium
94	Pu Plutonium
95	Am Americium
96	Cm Curium
97	Bk Berkelium
98	Cf Californium
99	Es Einsteinium
100	Fm Fermium
101	Md Mendelevium
102	No Nobelium
103	Lr Lawrencium

Elements with atomic numbers 112 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



2014
FORM V
ANNUAL EXAMINATION

General Instructions

- Write your class and candidate number in the space provided.
- Attempt all questions 1 – 13
- Use a blue or black pen
- Select the alternative A, B, C, or D that best answers the question.
- Fill in the response circle completely.

Class

Name

Chemistry
Part A
ANSWER SHEET

1. ☒ (A) ☐ (B) ☐ (C) ☐ (D)
2. ☐ (A) ☐ (B) ☐ (C) ☒ (D)
3. ☐ (A) ☐ (B) ☒ (C) ☐ (D)
4. ☐ (A) ☒ (B) ☐ (C) ☐ (D)
5. ☐ (A) ☐ (B) ☒ (C) ☐ (D)
6. ☒ (A) ☐ (B) ☐ (C) ☐ (D)
7. ☐ (A) ☐ (B) ☐ (C) ☒ (D)
8. ☐ (A) ☐ (B) ☒ (C) ☐ (D)
9. ☐ (A) ☒ (B) ☐ (C) ☐ (D)
10. ☐ (A) ☒ (B) ☐ (C) ☐ (D)
11. ☐ (A) ☐ (B) ☒ (C) ☐ (D)
12. ☐ (A) ☐ (B) ☐ (C) ☒ (D)
13. ☐ (A) ☒ (B) ☐ (C) ☐ (D)

Part B**Total marks (74)****Attempt ALL Questions**

TOTAL
= 19

Master's initials

Name

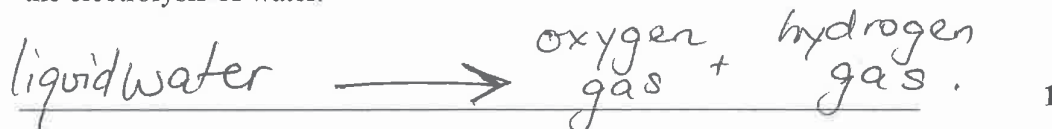
Answer the questions in the spaces provided.

Show **all** relevant working in questions involving calculations.**Question 14 (2 marks)****Marks**

Write word equations for;

NOTE: Boys still chose to write balanced chemical equations. (1 mark maximum if correct).

- (a) the electrolysis of water.



- (b) photosynthesis.

**Question 15 (3 marks)**

Write balanced chemical equations for;

- (a) the decomposition of silver chloride.



states not penalised here.

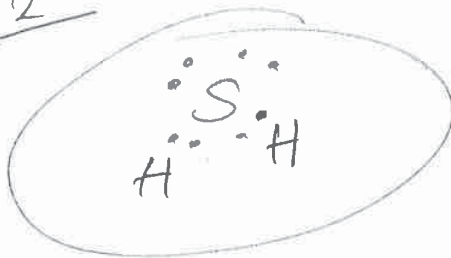
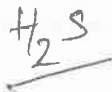
Boys still don't know chlorine is diatomic

- (b) the reaction between
- solutions
- of barium hydroxide and copper(II) sulfate.

NOTE: TWO solid products!

Question 16 (3 marks)**Marks**Draw Lewis electron dot diagrams for each of the following substances.

(a) hydrogen sulfide



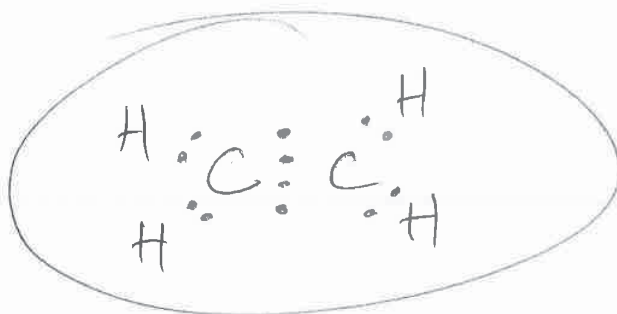
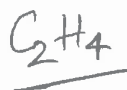
1

(b) nitrogen trifluoride



1

(c) ethene



1

Question 17 (4 marks)

Marks

The table below summarises some properties of Substance Y and Substance Z. Use the information to answer the following questions.

Substance Y	Substance Z
white solid that melts at 801 °C to form colourless liquid	grey solid that melts at 419 °C and boils at 906 °C
does not conduct electricity in solid state, but is a good conductor when liquid	conducts electricity in both solid and liquid states

- (a) Describe the structure and bonding of Substance Z at 25 °C.

(1 mark) 3D lattice structure 2
→ cations attracted to delocalised electrons. (1 mark)

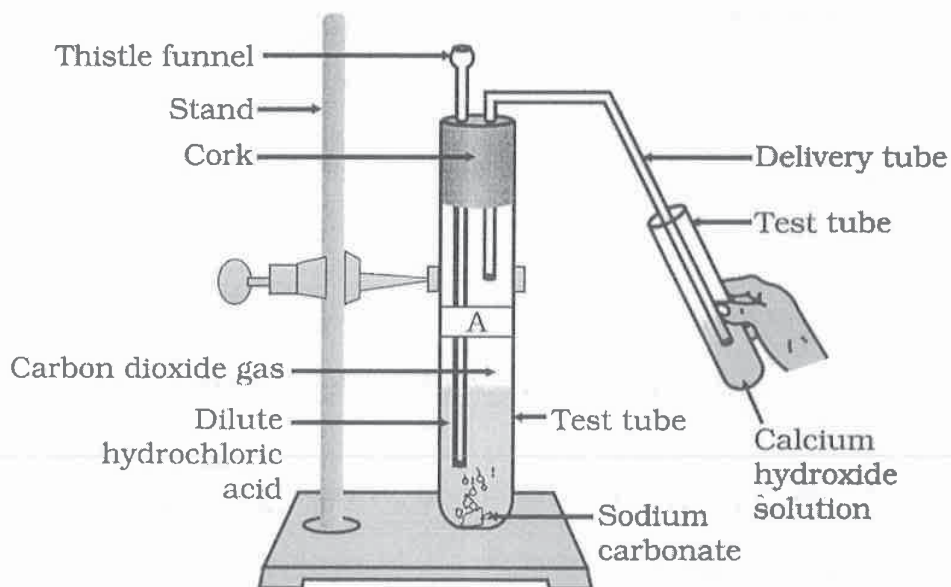
- (b) Predict whether an aqueous solution of Substance Y would conduct electricity, and justify your answer in terms of the properties of Substance Y.

(1 mark) It will conduct as it is an ionic compound (note properties in table) 2
(1 mark) cations and anions will be free to move in solution and carry charge. (conduct)

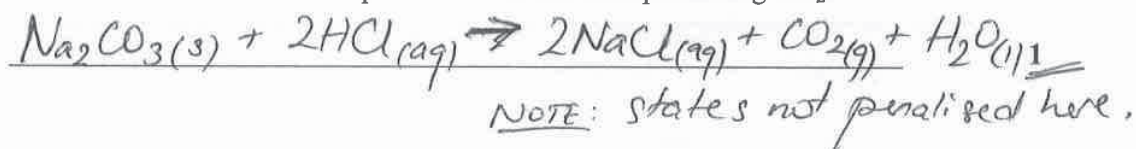
Question 18 (4 marks)

Marks

The diagram below shows an experimental set up for the production of carbon dioxide.



- (a) Write a balanced chemical equation for the reaction producing CO_2 .



- (b) If 2.50 g of solid sodium carbonate was reacted with excess hydrochloric acid, calculate the **mass and volume** of carbon dioxide evolved, measured at 100 kPa and 25 °C.

(1 mark) $n(\text{Na}_2\text{CO}_3) = \frac{2.5\text{g}}{106} = \boxed{0.024 \text{ moles}}$ 3

$\therefore$ 0.024 moles of $\text{CO}_2(\text{g})$ produced (1:1 mole ratio)

(1 mark) $0.024 \times \overset{\text{FW } \text{CO}_2}{(44)} = \boxed{1.04\text{g}}$ of $\text{CO}_2(\text{g})$ produced

(1 mark) $0.024 \times 24.79 = \boxed{0.58\text{L}}$ of $\text{CO}_2(\text{g})$ produced

NOTE: Carry over error considered if original equation incorrect and/or original number of moles miscalculated. EJS

Question 19 (3 marks)

Explain why different isotopes of an element have the same chemical properties, but different relative atomic masses.

(1 mark) Decent definition of isotope.

3

same element, but with a different
number of neutrons

—most mention protons/electrons the same

(1 mark) Neutron number affects the atomic mass

(1 mark) Electrons (valance/total) affect the
chemical properties (together with protons)

NOTE: must explicitly state why
chemical properties are the same

Not just — "neutrons are not
responsible for chemical
properties "

many answers!

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Question 20 (3 marks)

Marks

The history of metal use has been intricately interwoven with the evolution of society and civilisations throughout our history. Explain, citing examples, the general trend in technological developments and access to energy with regards to mankind's ability to extract metals from their ore.

Increased access to energy or better mining techniques / meant a greater variety of metals could be extracted (1)

3

Specific example of named metal and how technique which has altered its extraction (1)

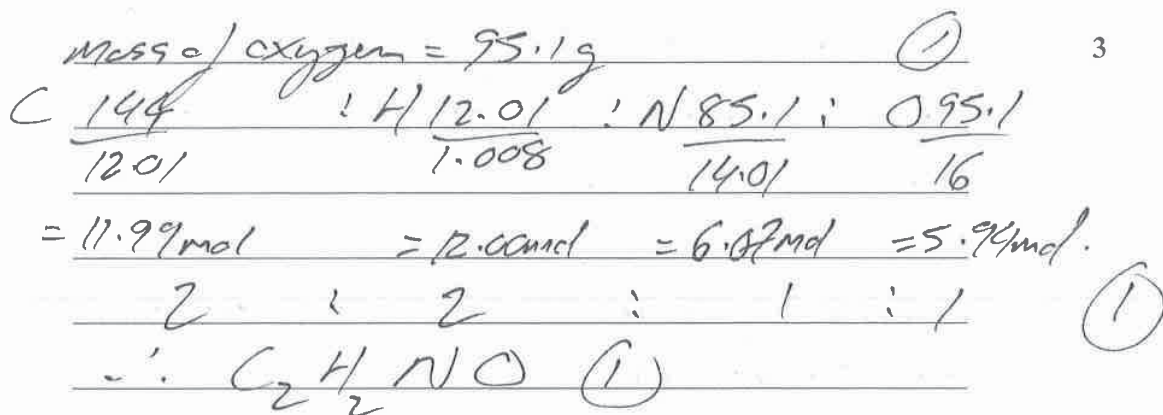
Increased access to energy has allowed us to purify increasingly reactive metals which requires breaking increasingly strong bonds.

Question 21 (5 marks)

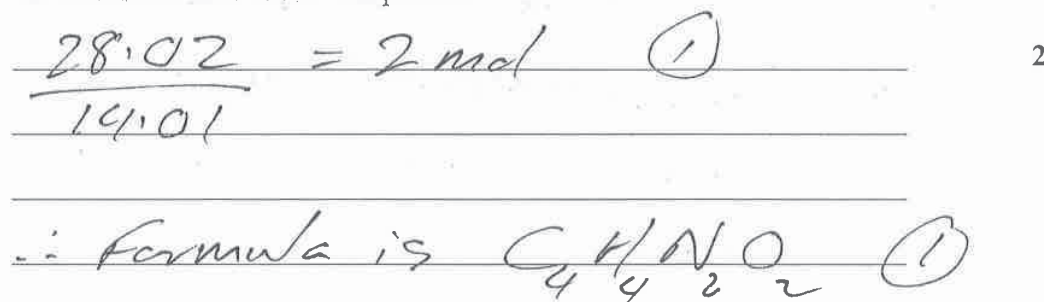
Marks

The compound uracil is one of the bases found in RNA (ribonucleic acid). A 336.3 g sample of uracil was broken down into its constituent elements and found to contain 144.0 g of carbon, 12.1 g of hydrogen, 85.1 g of nitrogen, with the remaining mass being oxygen.

- (a) Calculate the molar ratio of the elements in this compound and thus state its empirical formula.



- (b) If one mole of uracil contains 28.02 g of nitrogen, calculate the molecular formula of this compound.



Question 22 (6 marks)

Marks

The metal activity series can be experimentally derived using the reactions of metals with acids.

- (a) Redox reactions involve the transfer of electrons. Use a balanced chemical equation and the relevant half equations to explain why the reaction of zinc with aqueous sulfuric acid can be considered to be a redox reaction.



3



Explanation - is transfer of electrons (1)
OR oxidation of Zn & reduction of H^+

- (b) Not all metals react with acids. Explain why some metals will react with an acid (such as aqueous sulfuric acid), whilst others will not.

* Basic description of reaction including examples of metals that will react and those which will not. (1) 3

* Discussion of the relative pull of nucleus on the valence electrons (1)

* Relating pull of metal nucleus on electrons to pull of hydrogen nucleus on electrons & therefore ability of metal to react with acid (1)

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Master's initialsName**Question 23** (2 marks)**Marks**

Mendeleev is universally acknowledged as having put together the first functioning Periodic Table. Outline two decisions he made which allowed him to succeed in putting together a functioning Periodic Table when others had failed.

- Each work ① mark.
need 2 right none wrong for 2 marks.
- Organised by mass AND reactivity/properties.
 - Properties and reactivity took preference so swapped some elements eg. Te & I
 - left gaps for elements predicted to that he predicted were still to be found
- 2

Marks

Question 24 (5 marks)

The boiling points for Group V hydrides are shown in the table below.

Hydride	Boiling point ($^{\circ}\text{C}$)
ammonia (NH_3)	-33
phosphine (PH_3)	-88
arsine (AsH_3)	-55
stibine (SbH_3)	-17
bismuthine (BiH_3)	22

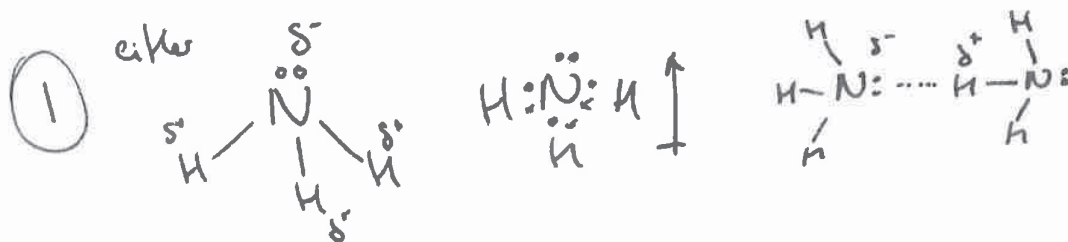
- (a) Explain the general trend in boiling points observed for phosphine to bismuthine.

• boiling point increases as mass increases or down the group.
 • As you go down mass & number of electrons increases $\therefore$ dispersion forces increase.

2

- (b) Using a diagram, explain why the boiling point of ammonia does not follow the same general trend as phosphine to bismuthine.

3



① Ammonia has hydrogen bonding which is caused by a highly polar bond due to a large difference in electronegativity.
 ① Hydrogen bonding much stronger than dipole-dipole or dispersion $\therefore$ requires more energy to disrupt.

Question 25 (5 marks)

Marks

At 25 °C, the maximum amount of sodium chloride that will dissolve in 100 mL of water is 35.9 g.

- (a) What is the concentration (in mol L⁻¹) of a saturated solution of sodium chloride at 25 °C?

$$\textcircled{1} \quad n = \frac{m}{M_r} \quad n = \frac{35.9}{58.44} = 0.61430527$$

2

$$\textcircled{1} \quad C = \frac{n}{V} = \frac{0.61430527}{0.1} = 6.14 \text{ mol L}^{-1}$$

- (b) 25.0 mL of a saturated solution of sodium chloride is diluted to 300 mL. Calculate the concentration (in mol L⁻¹) of the final solution.

1

$$C_1 V_1 = C_2 V_2 \quad C_2 = \frac{C_1 V_1}{V_2}$$

$$\text{as } n_1 = n_2 \quad = \frac{6.14 \times 0.025}{0.3} = 0.512 \text{ mol L}^{-1}$$

- (c) 40.0 g of sodium chloride is added to 100 mL of deionised water at 25 °C. Using this solution as an example, describe the movement of solute particles in a saturated solution.

$\textcircled{1}$ Dynamic equilibrium.

2

$\textcircled{1}$ Rate of precipitation equals the rate of dissolution.

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Question 26 (4 marks)

Marks

When 4.000 g of potassium hydroxide is dissolved in 100.00 g of water, the temperature of the water increases from 25.00 °C to 34.83 °C.

- (a) Calculate the chemical amount of potassium hydroxide that was used.

$$0.07129 \text{ mol}$$

1

- (b) Calculate the molar enthalpy of solution ($\Delta_{\text{sol}}H^\circ$, in kJ mol^{-1}) for potassium hydroxide.

3

$$\begin{aligned} q &= m C \Delta T \\ \textcircled{1} \quad &= 100 \text{ g} \times 4.18 \text{ J g}^{-1} \text{ K}^{-1} \times 9.83 \text{ K} \\ &= 4108.94 \text{ J} = 4.10894 \text{ kJ} \end{aligned}$$

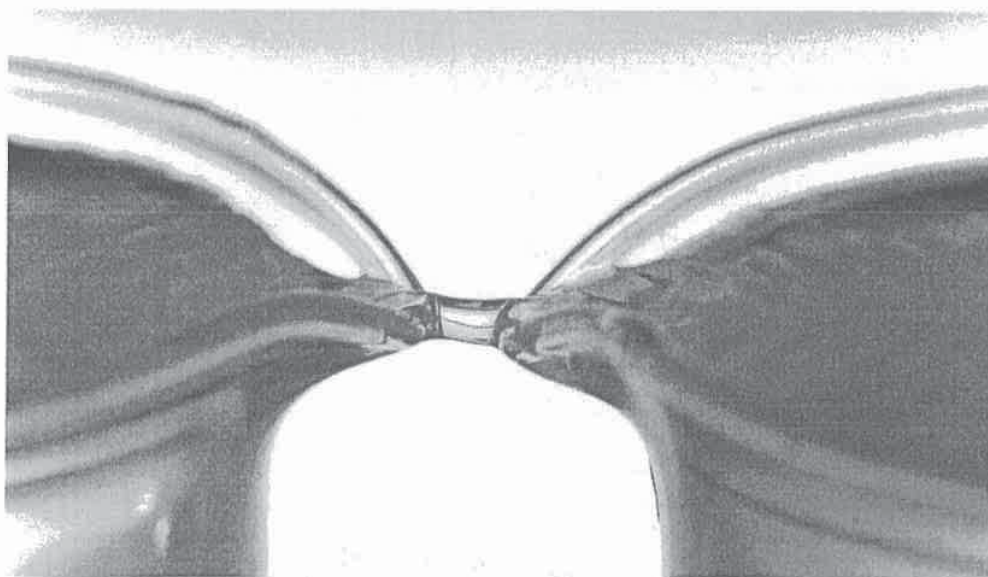
$$\begin{aligned} \Delta H &= \frac{-q}{n} = \frac{-4.10894 \text{ kJ}}{0.07129 \text{ mol}} \\ \textcircled{1} \quad &= -56.73 \text{ kJ mol}^{-1} \end{aligned}$$

$$\textcircled{1} \text{ sig figs } (4)$$

Question 27 (3 marks)

Marks

The image below shows a short "thread" of water suspended between two beakers.



Explain why it is possible to produce a stable "thread" of water as shown above.

3 marks - strong H-bonding between
- water molecules on surface have
unbalanced forces $\rightarrow$ high S.T.
- requires more force to break
thread to overcome tension / resists
an increase in S.A. etc 3

2 marks - good discussion of strong intermolecular
forces / cohesive forces, no link to
how high S.T. ~~cause~~ allows thread

1 mark - no explanation, only unrelated
statements.

Question 28 (2 marks)

A hydrocarbon will burn completely once its ignition temperature is reached, even though only a fraction of its molecules will have enough energy to react. Explain this observation.

① explanation of ignition temp

2

① exothermic combustion releases heat so more molecules have enough energy to overcome E_a .

Question 29 (3 marks)

Analyse the impact of increasing the temperature on reaction rate.

① more particles have $K.E. > E_a$
so ~~more~~ higher % successful collisions

3

① more KE so more collisions overall

① $\therefore$ The effect of $\uparrow T$ is $\uparrow$ rate.

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Question 30 (6 marks)

Marks

Carbon has a number of different allotropic forms, including diamond and graphite.

- (a) Compare the nature and geometry of the bonding in diamond and graphite.

1 Mark
1 Mark

Nature - diamond, localised ^{covalent} ~~tetrahedral~~, graphite ^{delocalised} ₂

Geometry - diamond tetrahedral, graphite - planar (trigonal)

Note Many boys launched into a discussion on structure.

If nothing else a mark given for identifying covalent bonds in common.

- (b) Under the right conditions, graphite can be converted into diamond.



The activation energy for this process is 120 kJ mol⁻¹.

- (i) Outline why the activation energy for this process is so high.

The carbon-carbon bonds in graphite are very strong and require a lot of energy to break.

- (ii) Given the above information, suggest what reaction conditions would be required to convert graphite into diamond.

High temperature

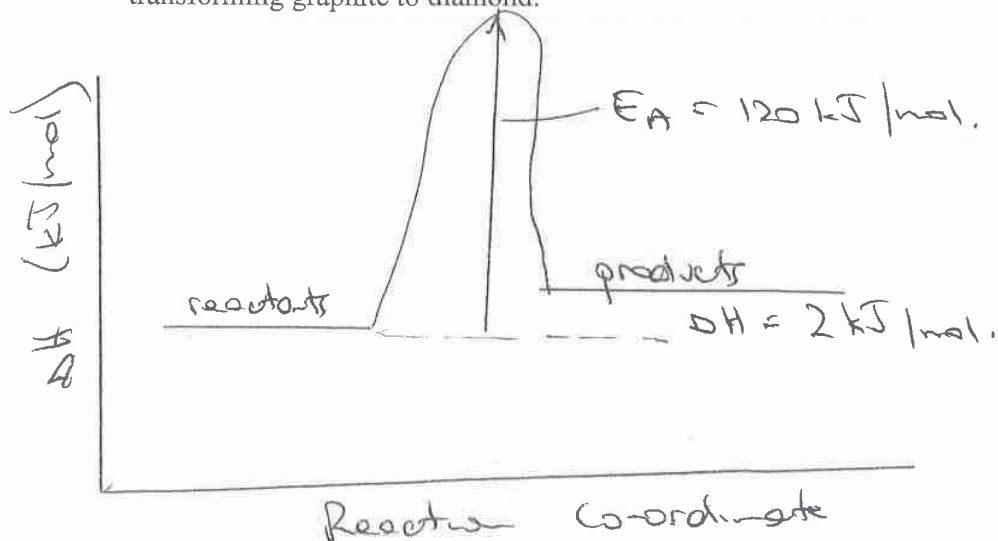
1

Question 30 continued on next page.

Question 30 continued.

Marks

- (ii) Draw a fully labelled energy profile diagram for the reaction transforming graphite to diamond.



Exothermic Reaction 0-Marks.

If ΔH in diagram is more than 50% of E_A then only one mark awarded (maximum)

Question 31 (4 marks)

Describe a reliable experiment that you could perform in the laboratory, using magnesium and hydrochloric acid, to investigate the effect of changing concentration on reaction rate. Include the expected result of this experiment.

- Mg mass (surface area) kept constant.
 - Concentration of HCl varied.
 - Identify Dependent variable to measure.
 - Duplicate experiment to increase reliability.
 - Sensible expected result.
- 4

All present	4 marks
missing 1	3 marks
missing 2	2 marks
missing 3	1 mark

note. For validity (not required) the volumes of the HCl should vary so that moles HCl remains constant!! Rarely considered.

Question 32 (3 marks)**Marks**

A BBQ heat bead, comprised of a mixture of graphite and clay, was placed in a container. An excess of oxygen gas was passed over the heat bead as it was being combusted. The graphite underwent complete combustion, but the clay remained unchanged. Masses of the container and its contents were documented during the experiment along with the mass of CO_2 produced.

Determine the percentage by mass of graphite in the BBQ heat bead using the information in the table below.

Item	Mass (kg)
reaction container	2.45
reaction container with heat beads before combustion	3.75
evolved CO_2	2.89

3

1 Mark.

moles CO_2 65.7 mol~~Mass~~ 1 Mark

mass graphite 788 g

1 Mark

% graphite 60%

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Question 33 (4 marks)

Marks

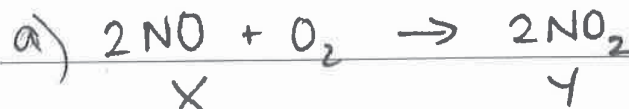
The following information refers to the reactions of an unknown compound, Compound Y. Compound Y is a molecular gas composed of nitrogen and oxygen. All gas volumes are measured at 25 °C and 100 kPa.

4

- Compound Y can be prepared by the reaction of another gas, Compound X, with oxygen. In this reaction, 1.00 L of Compound X reacts with 0.500 L of oxygen to produce 1.00 L of Compound Y.
- Compound Y will react with lead(II) oxide to produce Compound X and a white solid, Compound Z. In this reaction, 3.00 g of lead(II) oxide will react with 1.00 L of Compound Y, to produce 4.45 g of Compound Z and 0.333 L of Compound X. Compound Z is soluble in water.
- Compound Y has the interesting property that it can be used as a source of oxygen to promote combustion reactions. For example, it can be used as the oxygen source for the complete combustion of hydrocarbons, so that oxygen gas does not need to be used.

Showing all necessary logic, write balanced chemical equations for the three chemical reactions below. Ensure you clearly identify the formulae of Compounds X, Y and Z;

- The reaction of Compound X with oxygen to produce Compound Y.
- The reaction of Compound Y with lead(II) oxide to produce Compound Z and Compound X.
- The reaction of methane and Compound Y.



PTO

4 marks

- clear, logical process
- X, Y, Z clearly identified
- all equations correct
(c) complete combustion)

3 marks -

- as for 4 marks, but 1 error
e.g. not complete combustion
for (c)

2 marks

- at least one correct eq & an
 X/Y pair that matches data
or
- identifies Z and correct molar calcs

1 mark

- one equation that matches data
or
- correct molar calcs